EXP – 05

STOP AND WAIT PROGRAM

SERVER

```
import java.net.*;

public class StopWaitServer {
   public static void main(String[] args) {
      try {
         DatagramSocket serverSocket = new DatagramSocket(9000);
         System.out.println("Server started...");

         byte[] buffer = new byte[1024];

         // Receive sequence length from client
         DatagramPacket seqPacket =
               new DatagramPacket(buffer, buffer.length);
         serverSocket.receive(seqPacket);

         int sequenceLength = Integer.parseInt(
               new String(seqPacket.getData(), 0, seqPacket.getLength()));

         System.out.println("Sequence Length = " + sequenceLength);

         int ackCount = 0;

         while (ackCount < sequenceLength) {

            // Receive random number
```

```
        DatagramPacket receivePacket =
                new DatagramPacket(buffer, buffer.length);
        serverSocket.receive(receivePacket);

        int number = Integer.parseInt(
                new String(receivePacket.getData(), 0, receivePacket.getLength()));

        System.out.println("Received: " + number);

        int ack;
        if (number < 20) {
            ack = 0;
            System.out.println("ACK = 0 (Packet Lost)");
        } else {
            ack = 1;
            ackCount++;
            System.out.println("ACK = 1 (Accepted)  Count = " + ackCount);
        }

        // Send ACK to client
        String ackMsg = String.valueOf(ack);
        DatagramPacket ackPacket =
                new DatagramPacket(ackMsg.getBytes(), ackMsg.length(),
                        receivePacket.getAddress(), receivePacket.getPort());
        serverSocket.send(ackPacket);
    }

    System.out.println("All packets received successfully.");
    serverSocket.close();

} catch (Exception e) {
```

```
            e.printStackTrace();
        }
    }
}
```

CLIENT

```
import java.net.*;
import java.util.Random;

public class StopWaitClient {
    public static void main(String[] args) {
        try {
            DatagramSocket clientSocket = new DatagramSocket();
            InetAddress serverAddress = InetAddress.getLocalHost();
            byte[] buffer = new byte[1024];

            // Send sequence length
            int sequenceLength = 10;
            String seq = String.valueOf(sequenceLength);

            DatagramPacket seqPacket =
                    new DatagramPacket(seq.getBytes(), seq.length(),
                            serverAddress, 9000);
            clientSocket.send(seqPacket);

            Random random = new Random();
            int ackCount = 0;

            while (ackCount < sequenceLength) {

                // Generate random number (0–50)
```

```
        int number = random.nextInt(51);
        System.out.println("Sending: " + number);

        DatagramPacket sendPacket =
            new DatagramPacket(String.valueOf(number).getBytes(),
                String.valueOf(number).length(),
                serverAddress, 9000);
        clientSocket.send(sendPacket);

        // Receive ACK
        DatagramPacket ackPacket =
            new DatagramPacket(buffer, buffer.length);
        clientSocket.receive(ackPacket);

        int ack = Integer.parseInt(
            new String(ackPacket.getData(), 0, ackPacket.getLength()));

        if (ack == 1) {
            ackCount++;
            System.out.println("ACK = 1  Success Count = " + ackCount);
        } else {
            System.out.println("ACK = 0  Retransmitting");
        }
    }

    System.out.println("Transmission completed.");
    clientSocket.close();

} catch (Exception e) {
    e.printStackTrace();
}
```

```
    }

}
```

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>cd..

C:\Users>cd..

C:\>D:

D:\>cd share

D:\Share>cd A27

D:\Share\A27>javac StopWaitServer.java

D:\Share\A27>java StopWaitServer
Server started...
Sequence Length = 5
Received: 13
ACK = 0 (Packet Lost)
Received: 25
ACK = 1 (Accepted)  Count = 1
Received: 17
ACK = 0 (Packet Lost)
Received: 11
ACK = 0 (Packet Lost)
Received: 23
ACK = 1 (Accepted)  Count = 2
Received: 15
ACK = 0 (Packet Lost)
Received: 48
ACK = 1 (Accepted)  Count = 3
Received: 43
ACK = 1 (Accepted)  Count = 4
Received: 49
ACK = 1 (Accepted)  Count = 5
All packets received successfully.

D:\Share\A27>javac StopWaitServer.java

D:\Share\A27>java StopWaitServer
Server started...
Sequence Length = 10
Received: 27
```

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>cd..

C:\Users>cd..

C:\>D:

D:\>cd share

D:\Share>cd A27

D:\Share\A27>javac StopWaitClient.java

D:\Share\A27>java StopWaitClient
Sending: 13
ACK = 0  Retransmitting
Sending: 25
ACK = 1  Success Count = 1
Sending: 17
ACK = 0  Retransmitting
Sending: 11
ACK = 0  Retransmitting
Sending: 23
ACK = 1  Success Count = 2
Sending: 15
ACK = 0  Retransmitting
Sending: 48
ACK = 1  Success Count = 3
Sending: 43
ACK = 1  Success Count = 4
Sending: 49
ACK = 1  Success Count = 5
Transmission completed.

D:\Share\A27>javac StopWaitClient.java

D:\Share\A27>java StopWaitClient
Sending: 0

D:\Share\A27>javac StopWaitClient.java

D:\Share\A27>java StopWaitClient
```